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# PROJECT ON

# EMPLOYEE MANAGEMENT SYSTEM DATA ANALYSIS USING SQL


**Project Associates**

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# Objectives of the Employee Management System

- To design and develop a relational database for managing employee information.
- To store employee details such as Employee ID, Name, Department, Salary, and Designation.
- To establish relationships between Employees, Departments, and other related tables.
- To perform CRUD operations (Create, Read, Update, Delete) efficiently using SQL.
- To retrieve and analyze employee data using JOIN, GROUP BY, and aggregate functions.
- To maintain data integrity using Primary Key and Foreign Key constraints.
- To improve accuracy, reduce manual work, and ensure organized data management.

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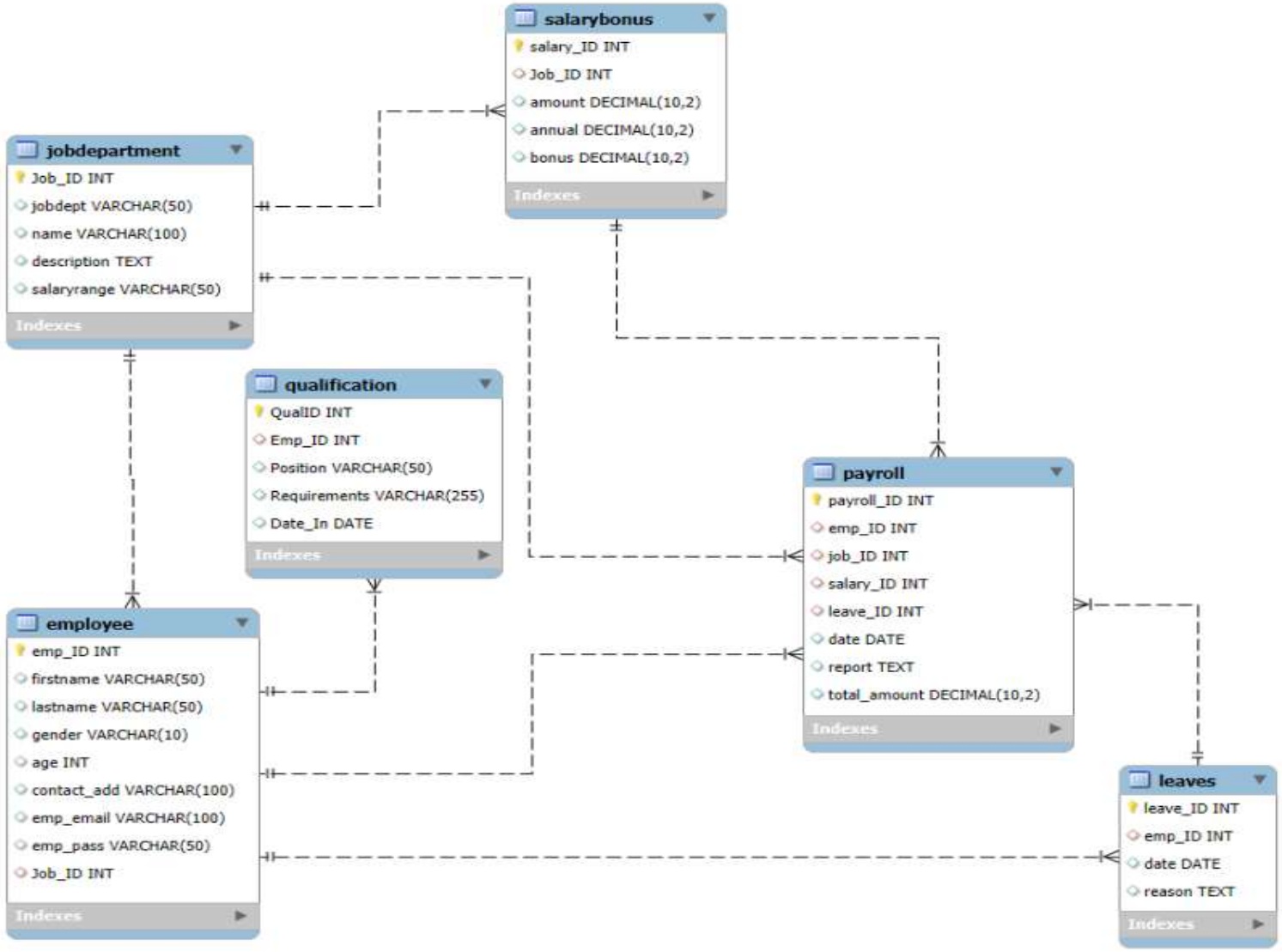
# ER Diagram

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# Tables

## 1. Job Department Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| job_id | int | NO | PRI | NULL | |
| jobdept | varchar(50) | YES | | NULL | |
| name | varchar(100) | YES | | NULL | |
| description | text | YES | | NULL | |
| salaryrange | varchar(50) | YES | | NULL | |

## 2. Salary Bonus Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| salary_id | int | NO | PRI | NULL | |
| job_id | int | YES | MUL | NULL | |
| amount | decimal(10,2) | YES | | NULL | |
| annual | decimal(10,2) | YES | | NULL | |
| bonus | decimal(10,2) | YES | | NULL | |

## 3. Leaves Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| leave_ID | int | NO | PRI | NULL | |
| emp_ID | int | YES | MUL | NULL | |
| date | date | YES | | NULL | |
| reason | text | YES | | NULL | |

## 4. Employee Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| emp_ID | int | NO | PRI | NULL | |
| firstname | varchar(50) | YES | | NULL | |
| lastname | varchar(50) | YES | | NULL | |
| gender | varchar(10) | YES | | NULL | |
| age | int | YES | | NULL | |
| contact_add | varchar(100) | YES | | NULL | |
| emp_email | varchar(100) | YES | UNI | NULL | |
| emp_pass | varchar(50) | YES | | NULL | |
| Job_ID | int | YES | MUL | NULL | |

## 5. Employee Qualification Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| QualId | int | NO | PRI | NULL | |
| emp_id | int | YES | MUL | NULL | |
| position | varchar(50) | YES | | NULL | |
| Requirements | varchar(255) | YES | | NULL | |
| Date_In | date | YES | | NULL | |

## 6. Payroll Table

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| payroll_ID | int | NO | PRI | NULL | |
| emp_ID | int | YES | MUL | NULL | |
| job_ID | int | YES | MUL | NULL | |
| salary_ID | int | YES | MUL | NULL | |
| leave_ID | int | YES | MUL | NULL | |
| date | date | YES | | NULL | |
| report | text | YES | | NULL | |
| total_amount | decimal(10,2) | YES | | NULL | |

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# Tables After Data Insertion

## 1. Job Department Table

| job_id | jobdept | name | description | salaryrange |
|---|---|---|---|---|
| 1 | Operations | Operations Engineer | Responsible for engineer duties in the operation... | $70889 - $91502 |
| 2 | Finance | Finance Consultant | Responsible for consultant duties in the finance ... | $45757 - $102274 |
| 3 | IT | IT Coordinator | Responsible for coordinator duties in the IT dep... | $32637 - $118326 |
| 4 | Marketing | Marketing Engineer | Responsible for engineer duties in the marketin... | $48936 - $145464 |
| 5 | Finance | Finance Specialist | Responsible for specialist duties in the finance d... | $67562 - $99403 |
| 6 | Engineering | Engineering Executive | Responsible for executive duties in the enginee... | $66000 - $125000 |
| 7 | Sales | Sales Manager | Responsible for managing the sales department. | $75000 - $130000 |
| 8 | IT | IT Analyst | Responsible for analyzing IT systems in the IT d... | $48000 - $92000 |

## 2. Salary Bonus Table

| salary_id | job_id | amount | annual | bonus |
|---|---|---|---|---|
| 1 | 1 | 45000.00 | 540000.00 | 5000.00 |
| 2 | 2 | 55000.00 | 660000.00 | 7000.00 |
| 3 | 3 | 40000.00 | 480000.00 | 4000.00 |
| 4 | 4 | 48000.00 | 576000.00 | 6000.00 |
| 5 | 5 | 47000.00 | 564000.00 | 4500.00 |
| 6 | 6 | 65000.00 | 780000.00 | 10000.00 |
| 7 | 7 | 80000.00 | 960000.00 | 12000.00 |
| 8 | 8 | 42000.00 | 504000.00 | 3500.00 |

## 3. Employee Table

| emp_ID | firstname | lastname | gender | age | contact_add | emp_email | emp_pass | Job_ID |
|---|---|---|---|---|---|---|---|---|
| 1 | John | Doe | M | 30 | 123 Elm St, NY | john.doe@example.com | password123 | 1 |
| 2 | Jane | Smith | F | 28 | 456 Maple Ave, CA | jane.smith@example.com | password123 | 2 |
| 3 | Alice | Brown | F | 35 | 789 Oak St, TX | alice.brown@example.com | password123 | 3 |
| 4 | Bob | Johnson | M | 40 | 101 Pine St, FL | bob.johnson@example.com | password123 | 4 |
| 5 | Charlie | Williams | M | 33 | 202 Birch Rd, IL | charlie.williams@example.com | password123 | 5 |
| 6 | Dave | Miller | M | 29 | 303 Cedar Blvd, NY | dave.miller@example.com | password123 | 6 |
| 7 | Eve | Davis | F | 25 | 404 Maple Dr, TX | eve.davis@example.com | password123 | 7 |
| 8 | Frank | Wilson | M | 38 | 505 Pine Ave, FL | frank.wilson@example.com | password123 | 8 |

## 4. Leaves Table

| leave_ID | emp_ID | date | reason |
|---|---|---|---|
| 1 | 1 | 2024-01-15 | Medical Leave |
| 2 | 2 | 2024-10-02 | Personal Work |
| 3 | 3 | 2024-05-03 | Sick Leave |
| 4 | 4 | 2024-01-25 | Family Emergency |
| 5 | 5 | 2024-10-03 | Travel |
| 6 | 6 | 2024-02-15 | Medical Checkup |
| 7 | 7 | 2024-01-04 | Festival |
| 8 | 8 | 2024-10-04 | Vacation |

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## 5. Employee Qualification Table

| QualId | emp_id | position | Requirements | Date_In |
|---|---|---|---|---|
| 1 | 1 | HR Assistant | BBA Communication Skills | 2019-12-06 |
| 2 | 2 | HR Manager | MBA 5 Yrs Exp | 2018-03-20 |
| 3 | 3 | IT Support | B.Tech (IT) Troubleshooting | 2020-01-15 |
| 4 | 4 | Software Engineer | B.Tech (CS) Java/Python | 2021-04-01 |
| 5 | 5 | Data Analyst | B.Sc Stats Excel SQL | 2020-08-11 |
| 6 | 6 | Sr. Data Analyst | M.Sc Stats Power BI | 2017-11-20 |
| 7 | 7 | Marketing Executive | BBA Google Ads | 2021-05-10 |
| 8 | 8 | Marketing Manager | MBA Marketing 5 Yrs Exp | 2018-07-14 |

## 6. Payroll Table

| payroll_ID | emp_ID | job_ID | salary_ID | leave_ID | date | report | total_amount |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 2024-04-30 | April Payroll | 40500.00 |
| 2 | 2 | 2 | 2 | 2 | 2024-04-30 | April Payroll | 45500.00 |
| 3 | 3 | 3 | 3 | 3 | 2024-04-30 | April Payroll | 39500.00 |
| 4 | 4 | 4 | 4 | 4 | 2024-04-30 | April Payroll | 43500.00 |
| 5 | 5 | 5 | 5 | 5 | 2024-04-30 | April Payroll | 38500.00 |
| 6 | 6 | 6 | 6 | 6 | 2024-04-30 | April Payroll | 42500.00 |
| 7 | 7 | 7 | 7 | 7 | 2024-04-30 | April Payroll | 44500.00 |
| 8 | 8 | 8 | 8 | 8 | 2024-04-30 | April Payroll | 45500.00 |

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# Key Analysis Questions (Use Cases)

## 1. EMPLOYEE INSIGHTS

1.1 How many unique employees are currently in the system?

1.2 Which departments have the highest number of employees?

1.3 What is the average salary per department?

1.4 Who are the top 5 highest-paid employees?

1.5 What is the total salary expenditure across the company?

## 2. JOB ROLE AND DEPARTMENT ANALYSIS

2.1 How many different job roles exist in each department?

2.2 What is the average salary range per department?

2.3 Which job roles offer the highest salary?

2.4 Which departments have the highest total salary allocation?

## 3. QUALIFICATION AND SKILLS ANALYSIS

3.1 How many employees have at least one qualification listed?

3.2 Which positions require the most qualifications?

3.3 Which employees have the highest number of qualifications?

## 4. LEAVE AND ABSENCE PATTERNS

4.1 Which year had the most employees taking leaves?

4.2 What is the average number of leave days taken by its employees per department?

4.3 Which employees have taken the most leaves?

4.4 What is the total number of leave days taken company-wide?

4.5 How do leave days correlate with payroll amounts?

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## 5. PAYROLL AND COMPENSATION ANALYSIS

5.1 What is the total monthly payroll processed?

5.2 What is the average bonus given per department?

5.3 Which department receives the highest total bonuses?

5.4 What is the average value of total_amount after considering leave deductions?

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# 1. EMPLOYEE INSIGHTS

## 1.1 How Many Unique Employees are currently in the System?

```
select count(*) as total_employees from Employee;
```

**Output :**

| | total_employees |
|---|---|
| ▶ | 60 |

**Insight :**

- The system currently records **60 employees**, representing the total workforce managed within the database. This dataset enables further analysis of department distribution, salary expenditure, leave patterns, and payroll trends.

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## 1.2 Which departments have the highest number of employees?

```
select jobdept,total_employees
from (select jd.jobdept,COUNT(e.emp_ID) as total_employees,
    rank() over (order by count(e.emp_ID) desc) as dept_rank
    from Employee e
    join JobDepartment jd on e.Job_ID = jd.Job_ID
    group by jd.jobdept
) as dept_count
where dept_rank = 1;
```

### Output :

Result Grid | Filter Rows:

| jobdept | total_employees |
|---|---|
| Finance | 9 |
| IT | 9 |

### Insight :

The **Finance and IT departments have the highest number of employees (9 each)**, indicating that these departments require the largest workforce to support the organization's operational and technological functions.

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## 1.3 What Is the Average salary Per department ?

```sql
select jd.jobdept,avg(sb.amount) as avg_salary
from employee e
join JobDepartment jd on e.job_id = jd.job_id
join SalaryBonus sb on jd.job_id = sb.job_id
group by jd.jobdept
order by avg_salary desc;
```

## Output :

Result Grid | Filter Rows:

| jobdept | avg_salary |
|---|---|
| Legal | 84600.000000 |
| Engineering | 81142.857143 |
| Sales | 75428.571429 |
| Finance | 72333.333333 |
| IT | 70888.888889 |
| Operations | 68750.000000 |
| Marketing | 65625.000000 |
| HR | 62571.428571 |

## Insight :

The **Legal department has the highest average salary (84.6K)**, followed by **Engineering (81.1K)**, reflecting the need for specialized expertise in these fields. In contrast, **HR and Marketing have lower average salaries**, suggesting a different compensation structure based on role specialization and operational requirements.

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## 1.4 Who are the top 5 highest-paid employees?

```
select e.firstname,e.lastname,sb.amount
from employee e
join salarybonus as sb on e.job_id = sb.job_id
order by sb.amount desc
limit 5;
```

## Output :

Result Grid | Filter Rows:

| firstname | lastname | amount |
|---|---|---|
| Ingrid | Adams | 170000.00 |
| John | Baker | 160000.00 |
| Grake | Moor | 150000.00 |
| Hank | Wilson | 150000.00 |
| Kelly | Cooper | 140000.00 |

## Insight :

The top five highest-paid employees earn between **140,000 and 170,000**, with **Ingrid Adams receiving the highest salary of 170,000**. This indicates that senior or specialized roles in the organization command significantly higher compensation.

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## 1.5 What is the total salary expenditure across the company?

```
select sum(sb.amount) as Total_Salary_Expenditure
from  employee e
join salarybonus sb on e.job_id = sb.job_id;
```

## Output :

Result Grid | Filter Rows:

| | Total_Salary_Expenditure |
|---|---|
| ▶ | 4321000.00 |

## Insight :

The company's total salary expenditure is approximately **4.32 million**, indicating a significant investment in employee compensation. With 60 employees in the system, the estimated **average salary per employee is around 72K**, reflecting the organization's compensation structure across departments.

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# 2. JOB ROLE AND DEPARTMENT ANALYSIS

## 2.1 How many different job roles exist in each department?

```
select jobdept, COUNT(name) AS no_of_job_roles
from jobdepartment
group by jobdept
order by no_of_job_roles desc;
```

## Output :

Result Grid | Filter Rows:

| jobdept | no_of_job_roles |
|---|---|
| Finance | 9 |
| IT | 9 |
| Operations | 8 |
| Marketing | 8 |
| Engineering | 7 |
| Sales | 7 |
| HR | 7 |
| Legal | 5 |

## Insight :

The **Finance and IT departments have the highest role diversity with 9 job roles each**, indicating complex functional structures. In contrast, the **Legal department has only 5 roles**, reflecting a more specialized and streamlined team.

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## 2.2 What is the average salary per department?

```
select jd.jobdept,avg(sb.amount) as average_salary
from jobdepartment jd
join salarybonus sb on
jd.job_id = sb.job_id
group by jd.jobdept
order by average_salary desc;
```

## Output :

Result Grid | Filter Rows:

| jobdept | average_salary |
|---|---|
| Legal | 84600.000000 |
| Engineering | 81142.857143 |
| Sales | 75428.571429 |
| Finance | 72333.333333 |
| IT | 70888.888889 |
| Operations | 68750.000000 |
| Marketing | 65625.000000 |
| HR | 62571.428571 |

## Insight :

The **Legal and Engineering departments have the highest average salaries**, reflecting the need for specialized expertise. In contrast, **HR and Marketing have lower average salaries**, indicating different compensation structures based on role specialization and operational responsibilities.

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## 2.3 Which Job roles offer the highest salary

```
select jd.name as job_role,
sb.amount as salary
from JobDepartment jd
join SalaryBonus sb
on jd.job_id = sb.job_id
order by sb.amount desc;
```

## Output :

Result Grid | Filter Rows:

| job_names | Highest_Amount |
|---|---|
| Finance Director | 170000.00 |
| Engineering Director | 160000.00 |
| Marketing Director | 150000.00 |
| IT Director | 150000.00 |
| Sales Director | 140000.00 |
| Operations Director | 135000.00 |
| Senior Legal Consultant | 130000.00 |
| HR Director | 120000.00 |
| Senior IT Consultant | 120000.00 |
| Legal Manager | 115000.00 |

## Insight :

The **Finance Director role offers the highest salary (170,000)**, followed by other director-level positions such as Engineering and Marketing Directors. This indicates that leadership and strategic roles receive the highest compensation within the organization.

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## 2.4 Which departments have the highest total salary allocation?

```sql
SELECT jobdept, total_salary
FROM (
    SELECT jd.jobdept, SUM(sb.amount) AS total_salary,
    RANK() OVER (ORDER BY SUM(sb.amount) DESC) AS dept_rank
    FROM jobdepartment jd
    JOIN salarybonus sb ON jd.job_id = sb.job_id
    GROUP BY jd.jobdept
    ) ranked_dept
WHERE dept_rank = 1;
```

## Output :

Result Grid | Filter Rows:

| jobdept | total_salary |
|---|---|
| Finance | 651000.00 |

## Insight :

The **Finance department has the highest total salary allocation (651,000)**, indicating that the organization invests significantly in financial leadership and expertise to manage budgeting, planning, and strategic financial operations.

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# 3. QUALIFICATION AND SKILLS ANALYSIS

## 3.1 How many employees have at least one qualification listed?

```
SELECT COUNT(DISTINCT emp_ID) AS employees_with_qualification
FROM Qualification;
```

## Output :

Result Grid | Filter Rows:

| | employees_with_qualification |
|---|---|
| ▶ | 60 |

## Insight :

All **60 employees in the system have at least one qualification recorded**, indicating a fully qualified workforce and suggesting that the organization prioritizes education and professional credentials during recruitment.

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## 3.2 Which positions require the most qualifications?

```sql
SELECT Position, COUNT(*) AS total_qualifications
FROM Qualification
GROUP BY Position
ORDER BY total_qualifications DESC;
```

## Output :

Result Grid | Filter Rows:

| Position | total_qualifications |
|---|---|
| HR Assistant | 1 |
| HR Manager | 1 |
| IT Support | 1 |
| Software Engineer | 1 |
| Data Analyst | 1 |
| Sr. Data Analyst | 1 |
| Marketing Executive | 1 |
| Marketing Manager | 1 |
| Finance Analyst | 1 |
| Sr. Finance Officer | 1 |

## Insight :

Each job position in the dataset has **exactly one qualification requirement**, indicating a standardized and clearly defined qualification structure for roles within the organization.

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## 3.3 Which employees have the highest number of qualifications?

```sql
SELECT emp_ID, firstname, lastname, total_qualifications
FROM (
        SELECT e.emp_ID, e.firstname, e.lastname,
            COUNT(q.QualID) AS total_qualifications,
            RANK() OVER (ORDER BY COUNT(q.QualID) DESC) AS qual_rank
        FROM Employee e
        JOIN Qualification q ON e.emp_ID = q.Emp_ID
        GROUP BY e.emp_ID
    ) ranked_emp
WHERE qual_rank = 1;
```

### Output :

Result Grid | Filter Rows:

| emp_ID | firstname | lastname | total_qualifications |
|---|---|---|---|
| 1 | John | Doe | 1 |
| 2 | Jane | Smith | 1 |
| 3 | Alice | Brown | 1 |
| 4 | Bob | Johnson | 1 |
| 5 | Charlie | Williams | 1 |
| 6 | Dave | Miller | 1 |
| 7 | Eve | Davis | 1 |
| 8 | Frank | Wilson | 1 |
| 9 | Grace | Moore | 1 |
| 10 | Harry | Taylor | 1 |

### Insight :

All employees have **exactly one qualification recorded**, indicating a standardized qualification structure where each employee is associated with a single primary credential.

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# 4. LEAVE AND ABSENCE PATTERNS

## 4.1 Which year had the most employees taking leaves?

```
select year(date) as leave_year,
count(*) as total_leaves
from Leaves
group by year(date)
order by total_leaves desc;
```

### Output :

Result Grid | Filter Rows:

| | leave_year | total_leaves |
|---|---|---|
| ▶ | 2024 | 60 |

### Insight :

The year **2024 recorded the highest number of leaves with 60 leave entries**, indicating that all recorded leave activity in the dataset occurred during this year.

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# 4.2 What is the average number of leave days taken by its employees per department?

```sql
select jd.jobdept,
    avg(emp_leave.total_leaves) as avg_leave_days
from (  select emp_ID, COUNT(*) as total_leaves from Leaves
        group by emp_ID)as  emp_leave
join Employee e on emp_leave.emp_ID = e.emp_ID
join JobDepartment jd on e.Job_ID = jd.Job_ID
group by jd.jobdept
order by avg_leave_days desc;
```

## Output :

Result Grid | Filter Rows:

| jobdept | avg_leave_days |
|---|---|
| Operations | 1.0000 |
| Finance | 1.0000 |
| IT | 1.0000 |
| Marketing | 1.0000 |
| Engineering | 1.0000 |
| Sales | 1.0000 |
| HR | 1.0000 |
| Legal | 1.0000 |

## Insight :

The average number of leave days per employee is **1 across all departments**, indicating consistent leave utilization and balanced workforce attendance throughout the organization.

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## 4.3 Which employees have taken the most leaves?

```
SELECT e.emp_ID,e.firstname,e.lastname,
count(l.leave_ID) as total_leaves
from Leaves l
join Employee e on l.emp_ID = e.emp_ID
group by e.emp_ID
order by total_leaves desc;
```

## Output :

Result Grid | Filter Rows:

| emp_ID | firstname | lastname | total_leaves |
|---|---|---|---|
| 1 | John | Doe | 1 |
| 2 | Jane | Smith | 1 |
| 3 | Alice | Brown | 1 |
| 4 | Bob | Johnson | 1 |
| 5 | Charlie | Williams | 1 |
| 6 | Dave | Miller | 1 |
| 7 | Eve | Davis | 1 |
| 8 | Frank | Wilson | 1 |
| 9 | Grace | Moore | 1 |
| 10 | Harry | Taylor | 1 |

## Insight :

All employees have **exactly one recorded leave**, indicating a uniform leave distribution with no employee taking more leave than others in the dataset.

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## 4.4 What is the total number of leave days taken company-wide?

```
select count(*) as total_number_of_leaves
from Leaves;
```

### Output :

Result Grid | Filter Rows

| | total_number_of_leaves |
|---|---|
| ▶ | 60 |

### Insight :

Employees collectively took **60 leave days across the organization**, averaging **one leave day per employee**, indicating balanced leave utilization within the workforce.

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## 4.5 How do leave days correlate with payroll amounts?

```
SELECT e.emp_ID,
    COUNT(l.leave_ID) AS total_leaves,
    SUM(p.total_amount) AS total_salary
FROM Employee e
LEFT JOIN Leaves l ON e.emp_ID = l.emp_ID
LEFT JOIN Payroll p ON e.emp_ID = p.emp_ID
GROUP BY e.emp_ID;
```

## Output :

Result Grid | Filter Rows:

| emp_ID | total_leaves | total_salary |
|---|---|---|
| 1 | 1 | 40500.00 |
| 2 | 1 | 45500.00 |
| 3 | 1 | 39500.00 |
| 4 | 1 | 43500.00 |
| 5 | 1 | 38500.00 |
| 6 | 1 | 42500.00 |
| 7 | 1 | 44500.00 |
| 8 | 1 | 45500.00 |
| 9 | 1 | 40500.00 |
| 10 | 1 | 38500.00 |
| 11 | 1 | 49500.00 |
| 12 | 1 | 51500.00 |

## Insight :

All employees have **one leave day**, while payroll amounts vary across employees. This indicates that payroll differences are driven by **job roles and salary structures rather than leave deductions**, making it difficult to identify a correlation between leave days and payroll in this dataset.

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# 5. PAYROLL AND COMPENSATION ANALYSIS

## 5.1 What is the total monthly payroll processed?

```
select year(date) as year,
month(date) as month,
sum(total_amount) as total_monthly_payroll
from Payroll
group by year(date),month(date)
order by year,month;
```

## Output :

Result Grid | Filter Rows: | Export:

| year | month | total_monthly_payroll |
|---|---|---|
| 2024 | 4 | 2778000.00 |

## Insight :

The total payroll processed in **April 2024 was 2,778,000**, representing the company's monthly salary expenditure for employees during that period.

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## 5.2 What is the average bonus given per department?

```
select jd.jobdept,avg(sb.bonus) as average_bonus
from  jobdepartment jd
join salarybonus sb on jd.job_id = sb.job_id
group by jd.jobdept
order by average_bonus desc;
```

## Output :

Result Grid | Filter Rows:

| jobdept | average_bonus |
|---|---|
| Legal | 13300.000000 |
| Engineering | 12571.428571 |
| Sales | 11214.285714 |
| Finance | 10666.666667 |
| IT | 10444.444444 |
| Operations | 9687.500000 |
| Marketing | 9125.000000 |
| HR | 8171.428571 |

## Insight :

The **Legal department receives the highest average bonus (13.3K)**, followed by **Engineering (12.6K)**. In contrast, **HR and Marketing have lower average bonuses**, indicating differences in incentive structures across departments based on role specialization and responsibilities.

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## 5.3 Which department receives the highest total bonuses?

```
select jobdept, total_bonus
from (select jd.jobdept,sum(sb.bonus) AS total_bonus,
      rank() over(order by sum(sb.bonus) DESC) AS dept_rank
    from jobdepartment jd
    join salarybonus sb on jd.job_id = sb.job_id
    group by jd.jobdept
) ranked_dept
where dept_rank = 1;
```

## Output :

Result Grid | Filter Rows:

| jobdept | total_bonus |
|---|---|
| Finance | 96000.00 |

## Insight :

The **Finance department receives the highest total bonus allocation (96,000)**, indicating that employees in this department receive the largest share of incentive rewards within the organization.

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## 5.4 What is the average value of total_amount after considering leave deductions?

```
SELECT AVG(adjusted_salary) AS avg_adjusted_amount
FROM ( SELECT p.emp_ID,
            p.total_amount - COUNT(l.leave_ID) * 500 AS adjusted_salary
        FROM Payroll p
        LEFT JOIN Leaves l ON p.emp_ID = l.emp_ID
        GROUP BY p.emp_ID, p.total_amount
    ) salary_calc;
```

### Output :

Result Grid | Filter Rows:

| | avg_adjusted_amount |
|---|---|
| ▶ | 45800.000000 |

### Insight :

After considering leave deductions, the **average adjusted payroll amount is 45,800**, showing the average salary employees receive after accounting for leave-related deductions.

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# SQL Query Results & Insights

- To wrap things up, our data analysis doesn't just show us where we are; it highlights what we need to fix.
- On the business side, we need to review if our top-heavy compensation and flat bonus structures are actually driving the behaviors we want.
- On the technical side, this analysis exposed some major gaps in our data collection.
- Our leave tracker is currently generating uniform, constrained data, and our payroll records are limited to a single month.
- My primary recommendation is that IT and HR collaborate to clean up these data pipelines so our future reporting can be even more accurate.

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# Final business insights and recommendations

- To conclude, running the numbers is only half the job; the real value is in how we respond to them.
- From a strategic standpoint, we have an opportunity to modernize our compensation bands—specifically by making our bonus structures more performance-driven to engage our mid-level talent.
- From a technical and analytics standpoint, this project highlighted critical blind spots.
- Our leave tracking is capturing default constraints rather than real human behavior, and our payroll data lacks the historical depth needed for predictive forecasting.
- Fixing these data pipelines should be an immediate priority for our internal technical teams.
- If we look at cost efficiency, we see an interesting split in our HR strategy. We are maintaining large, operational teams in Finance and IT where the average compensation is tightly controlled in the $70,000 range.

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# Conclusion

- **Operational Investment vs. Premium Talent:** The organization concentrates its largest headcount and total payroll budgets in core infrastructure teams (Finance and IT), while paying the highest per-capita premiums for specialized roles (Legal and Engineering).

- **Top-Heavy Compensation Structure:** Financial rewards are strictly hierarchical, with Director-level positions exclusively dominating the top earnings and average bonuses acting as a rigid percentage of base pay rather than reflecting dynamic performance.

- **Critical Data Integrity Limitations:** The HR database exhibits severe synthetic constraints—specifically, an artificial flat-rate of exactly one leave day per employee and a lack of historical payroll data outside of April 2024.

- **Strategic Path Forward:** To transition from basic baseline reporting to advanced HR analytics, leadership must first audit the data ingestion pipelines to capture real-world variance, followed by a strategic review of mid-level compensation and performance metrics.

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# ANY QUESTIONS

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# Experience & Challenges

- Before we conclude, I want to briefly touch on the experience of working with this dataset and the challenges we encountered.
- On the positive side, the relational structure of the database was very clean, which made it straightforward to aggregate our core HR metrics.
- However, our biggest challenge was the data constraints.
- Because the attendance data assigned exactly one leave to every employee, and the payroll data was isolated to just April 2024, we were limited in our ability to run real-world correlation models or long-term forecasts.
- Recognizing these limitations is what drives our recommendation to expand our historical data ingestion moving forward.

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# THANK YOU

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